build_cross_validation_grid_candidates.R

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`build_cross_validation_grid_candidates`
Make Cross-Validation Grid Candidates

Description

Derives candidate spatial-grid widths from projected extent, location density, and a target number of locations per occupied cell.

Usage

```
build_cross_validation_grid_candidates(
  data_locations = NULL,
  target_locations_per_cell = 5,
  grid_size_multipliers = 2^base::seq(-2, 2)
)
```

Arguments

`data_locations` Location table returned by [build_cross_validation_location_table()] with finite ‘coord_x_km‘ and ‘coord_y_km‘ columns.

`target_locations_per_cell`
Single positive numeric target occupancy. Defaults to ‘5‘.

`grid_size_multipliers`
Numeric vector of unique positive multipliers applied to the density-derived baseline width. Defaults to ‘2 ^ seq(-2, 2)‘.

Details

The baseline width is the square root of projected bounding-box area divided by the expected occupied-cell count, where expected cells equal location count divided by target occupancy. Candidate widths therefore adapt to each spatial ID rather than using fixed kilometre widths by tier.

Value

Tibble with one row per candidate and columns describing candidate ID, cell width, baseline width, multiplier, location count, projected extents, area, and target occupancy.

Examples

```
data_locations <-
  tibble::tibble(
    coord_x_km = c(0, 0, 10, 10),
    coord_y_km = c(0, 10, 0, 10)
  )
build_cross_validation_grid_candidates(
  data_locations = data_locations,
  target_locations_per_cell = 1,
  grid_size_multipliers = c(0.5, 1, 2)
)
```

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